

Chapter 6

The Foreign Exchange Market

Chapter Outline

- Organization of the FX market
- Spot market
- Forward market

Organization of the FX Market

- Why foreign exchange markets exist
- Participants
- Global FX exchange

Why Foreign Exchange Markets Exist

- To facilitate purchases of goods and services between the home country and foreign country (i.e., imports and exports), currencies must be exchanged:
 - From \$ to €, € to \$, € to ¥, ¥ to €, and all other currency pairs.
- 2021 exports of goods and services for select countries as a percent of GDP (imports are roughly the same):

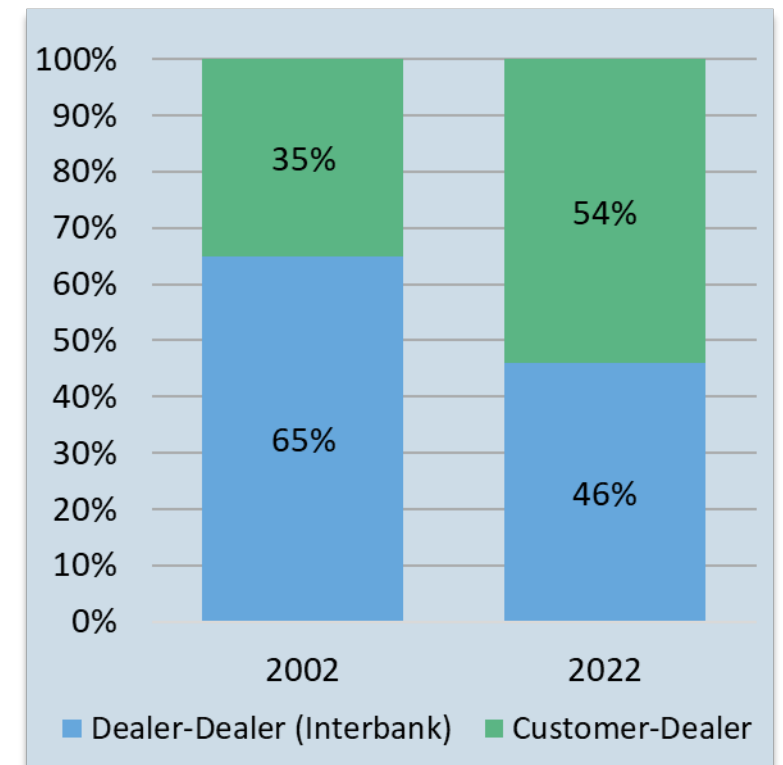
Country	Export % of GDP
US	11%
Japan	18.4%
Canada, France, Italy, UK	~30%
Germany	47%

- However, 95% of all FX transactions are not related to trade, but to cross-border purchases and sales of assets (international capital flows).

Participants

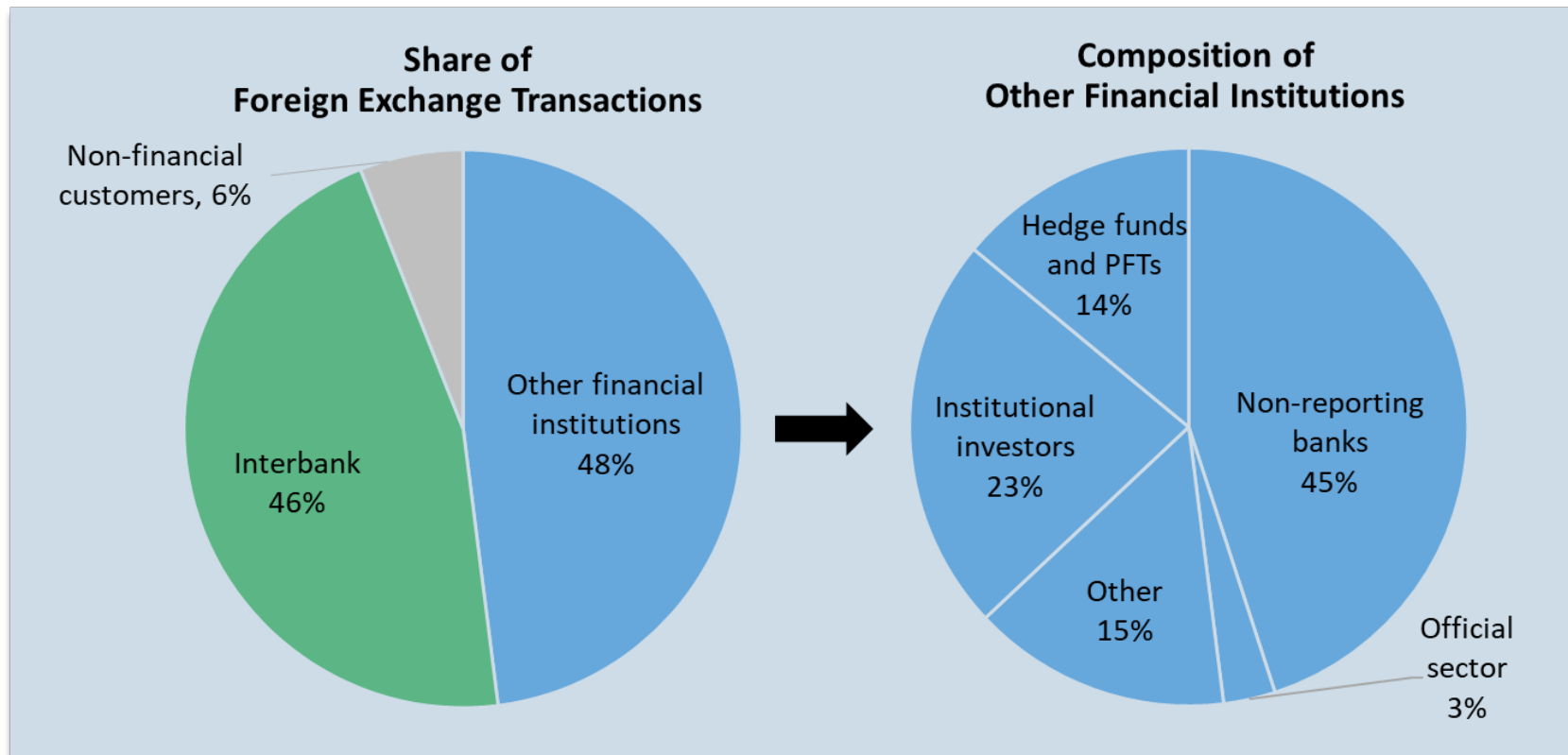
- Traditionally, most currency transactions were conducted **dealer to dealer** (interbank market), also called the wholesale market.
- Over the past 20 years, transactions are increasingly conducted **customer to dealer**.
- The interbank market is dominated by about 20 major banks.
 - The banks communicate and conduct trades through the **SWIFT** (Society For Worldwide Interbank Financial Telecommunications) electronic network of banks.
 - E.g., Goldman Sachs in New York can communicate with BNP Paribas in Paris to exchange USD for EUR using SWIFT.
- However, to find each other, banks often use an **FX broker**, who for a fee (1/32 of 1%), will match one bank with another.

Changing composition of transaction processing methods



Participants (2 of 3)

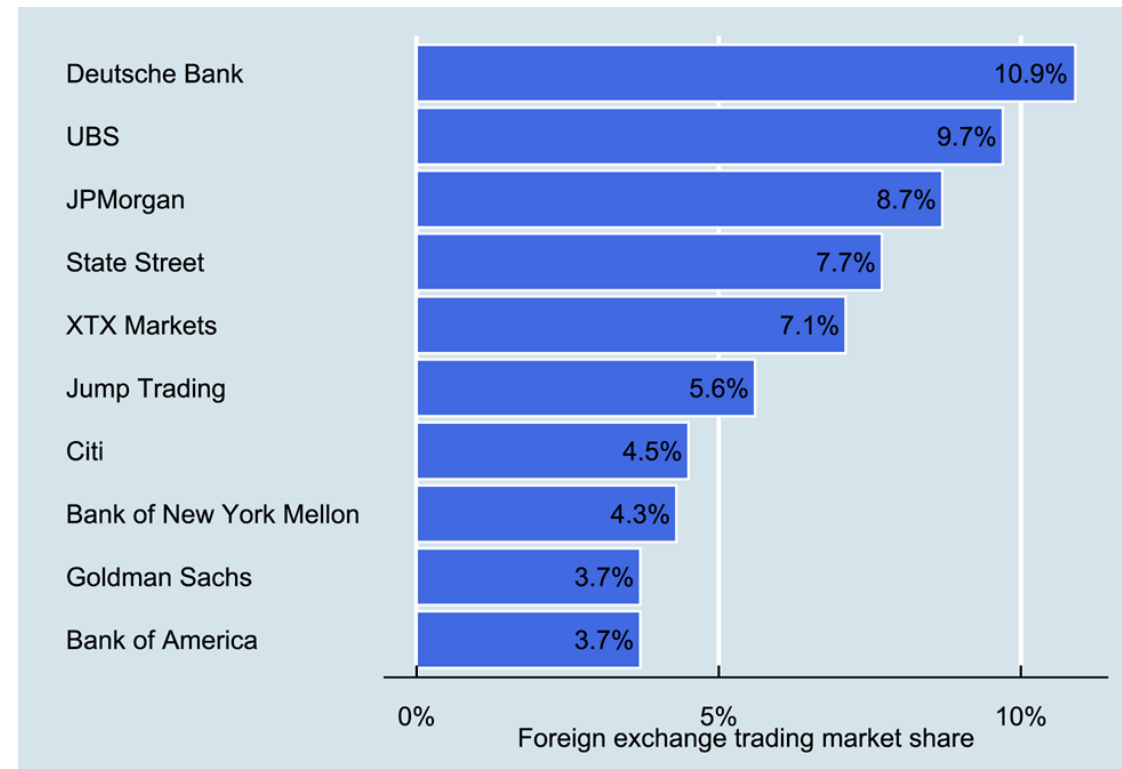
Composition of Foreign Exchange Market, 2022



1.2 Participants

- The 10 largest traders account for about 66% of all FX trades.
- For the first time, two are FX traders, not banks: XTX Markets (London) and Jump Trading (Chicago).
- American firms dominate the rankings.

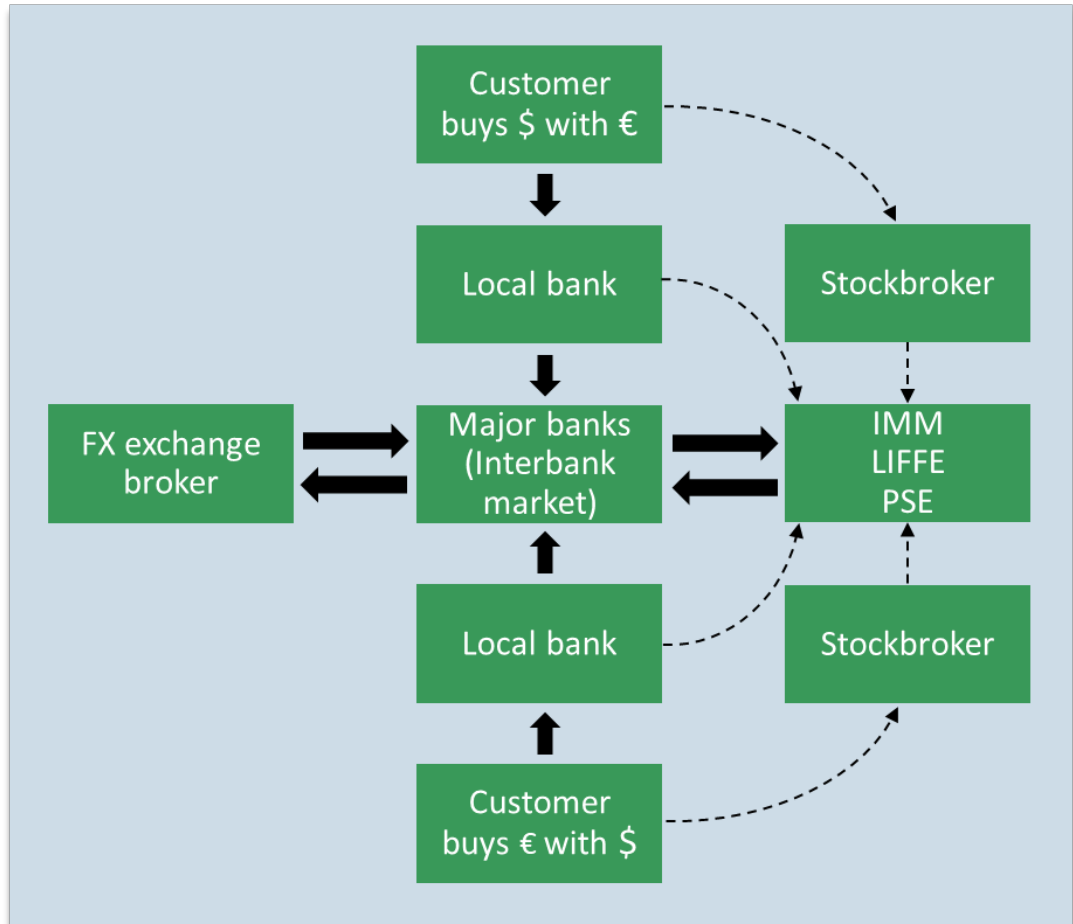
Top 10 foreign exchange traders by market share, 2022



Participants (3 of 3)

FX participant linkages

- The interbank market is at the core of the FX exchange ecosystem.
- Major banks sometimes use FX brokers to facilitate match-making.
- Retail customers go through local banks, which place orders with the major banks.
- Linkages to the currency futures and options are discussed in chapter 7.

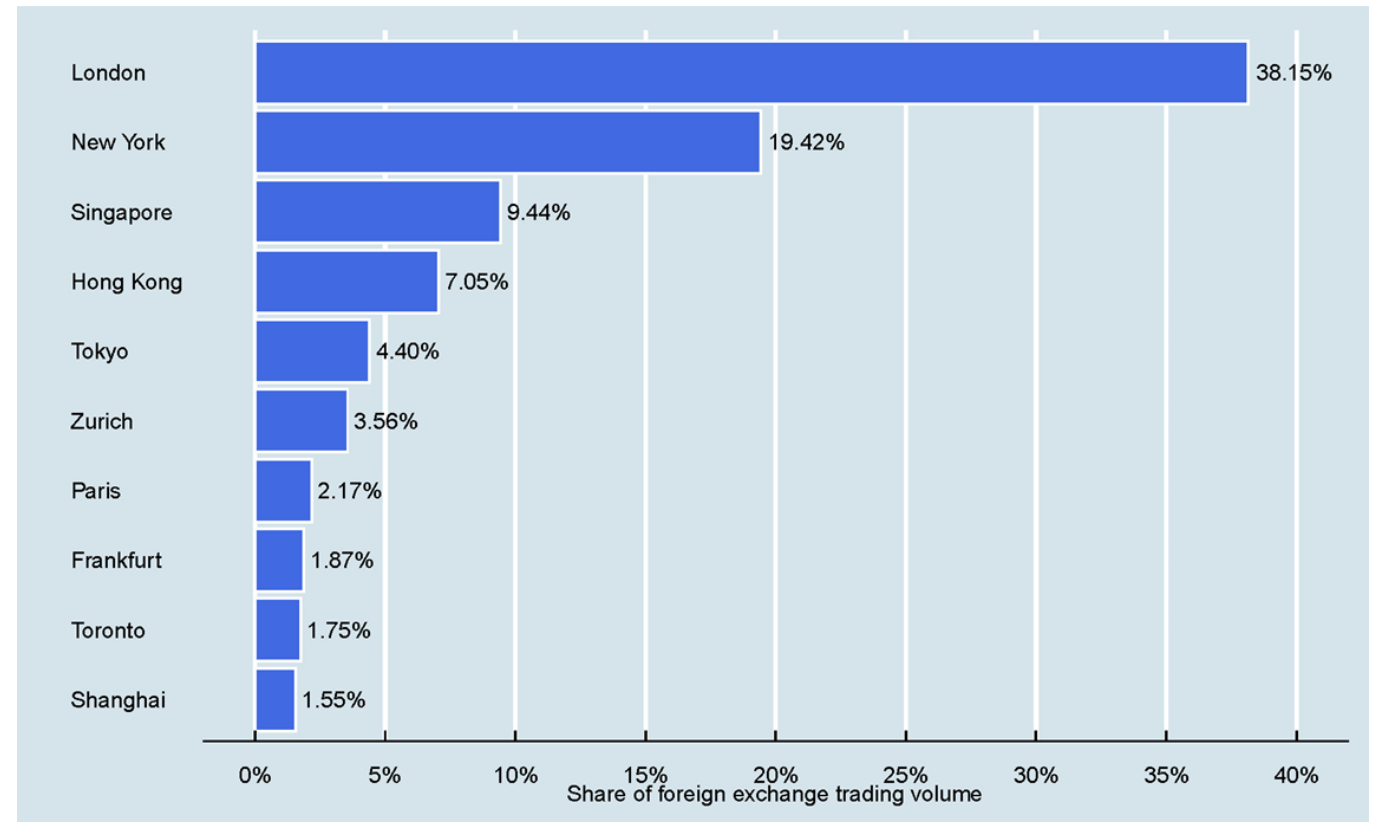


Global FX Exchange (1 of 4)

Global trading centers

- London, with more than 1/3 of all trading volume, is the largest FX financial center.
- The top 10 FX cities account for almost 90% of all trades.

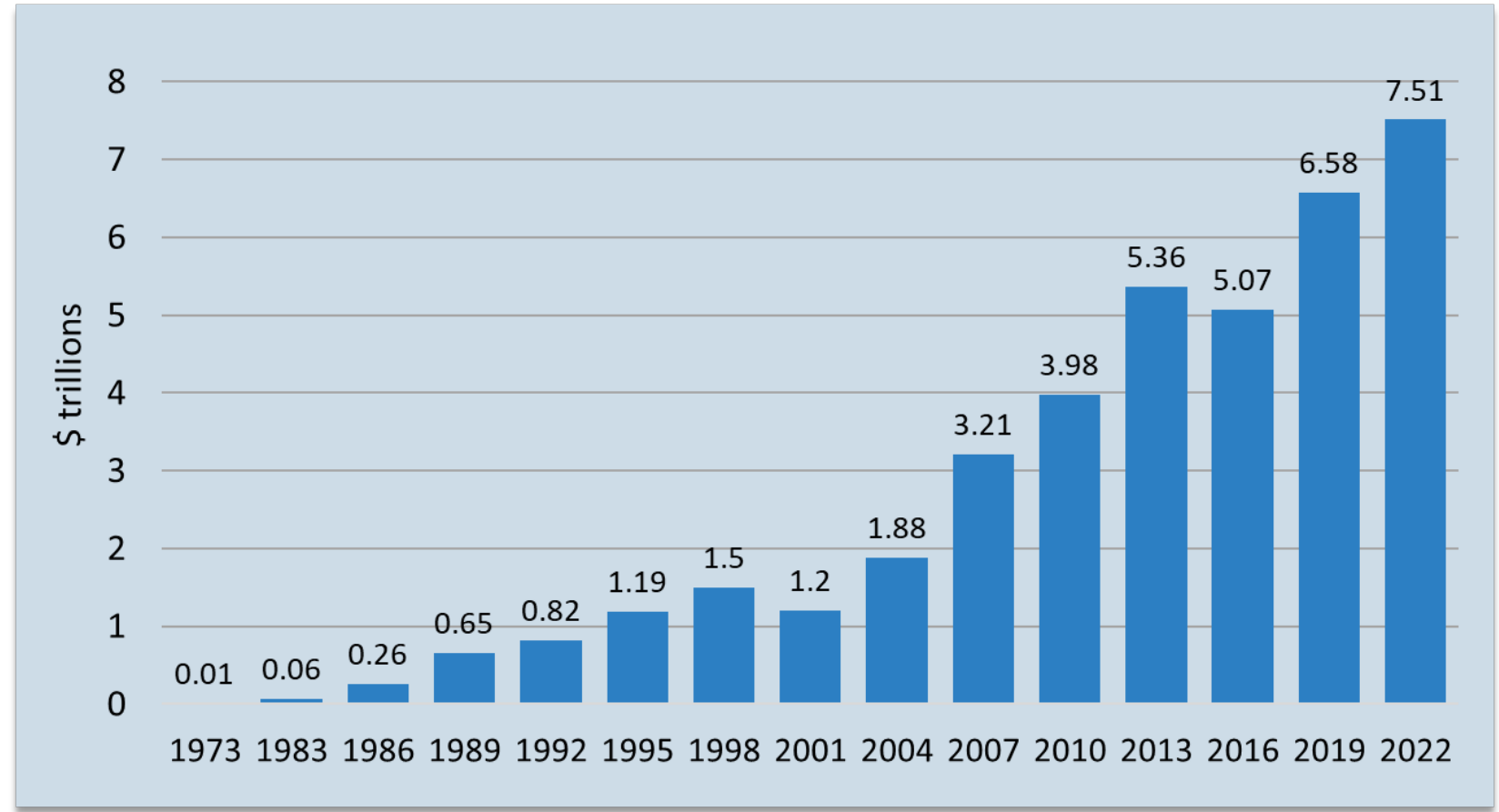
Top 10 FX centers by trading volume, 2022



Global FX Exchange (2 of 4)

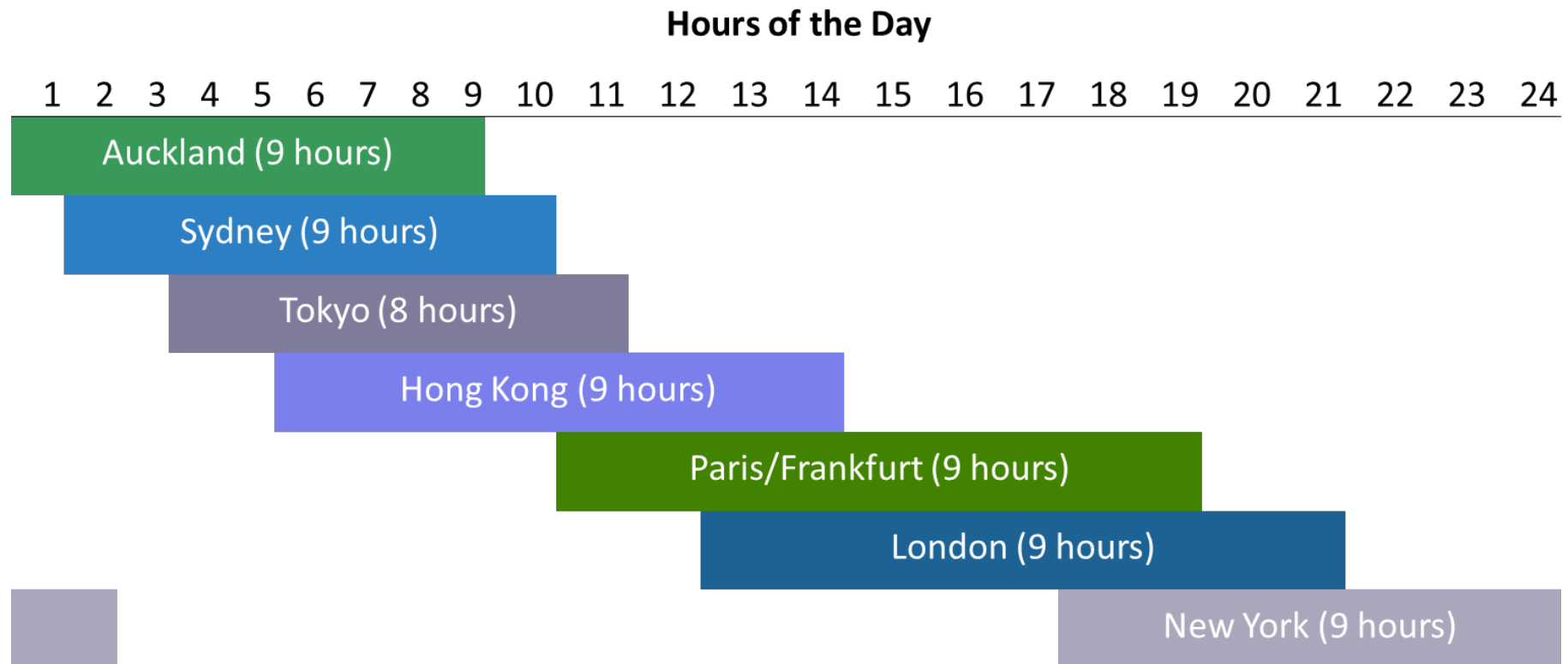
Daily volume

- The FX market is by far the largest financial market in the world.
- The daily volume of FX is 30 times daily world GDP and 13 times the average daily turnover of all equity stock markets put together.



Global FX Exchange (3 of 4)

Global trading hours

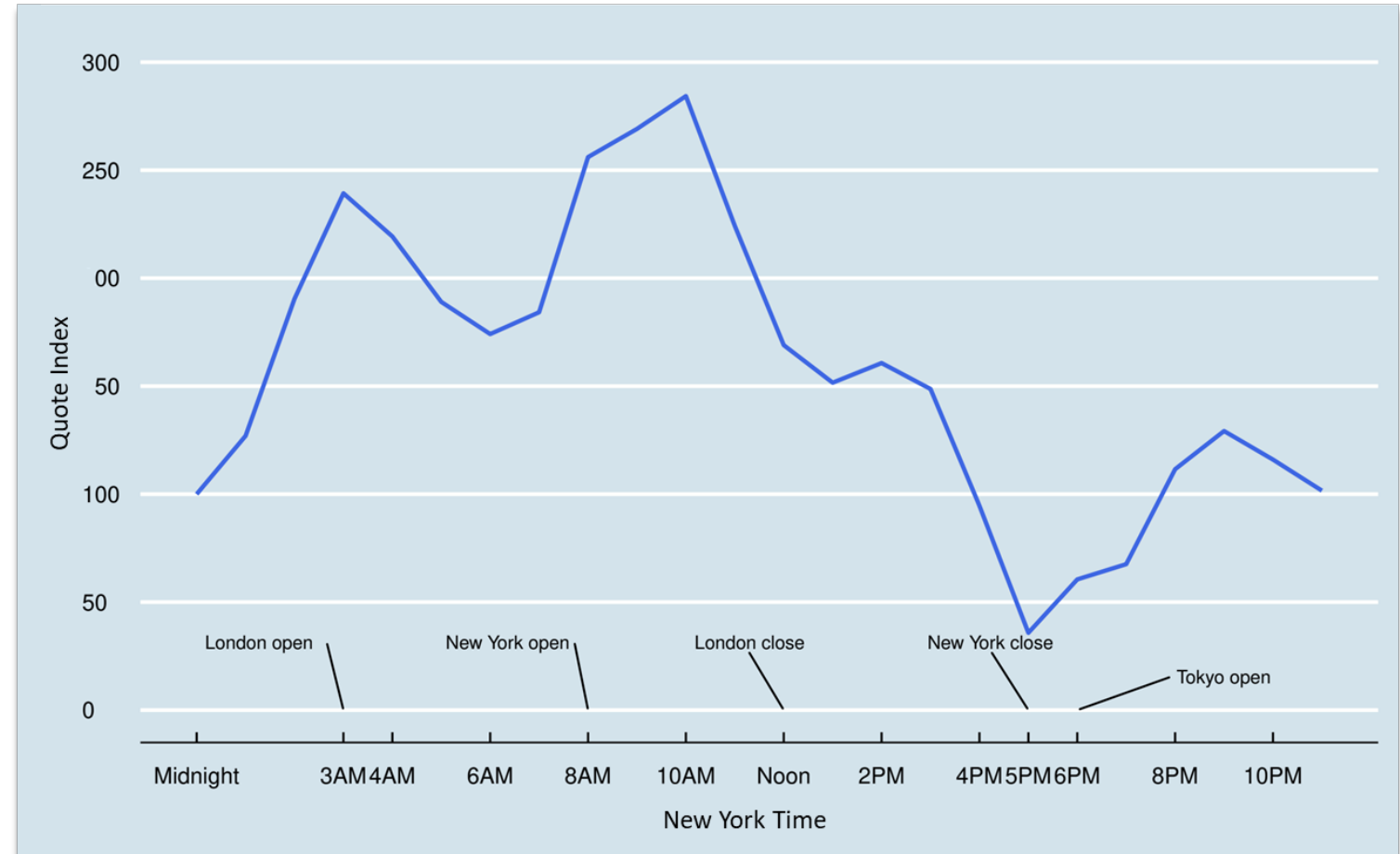


What are the hours of foreign exchange trading

Major market	Forex market timing (in UTC- Coordinated Universal Time)
Sydney	9:00 PM to 6:00 PM UTC
Tokyo	11:00 PM to 6:00 AM UTC
London	7:00 AM to 4:00 PM UTC
New York	12:00 PM to 9:00 PM UTC

1.3 Global FX Exchange

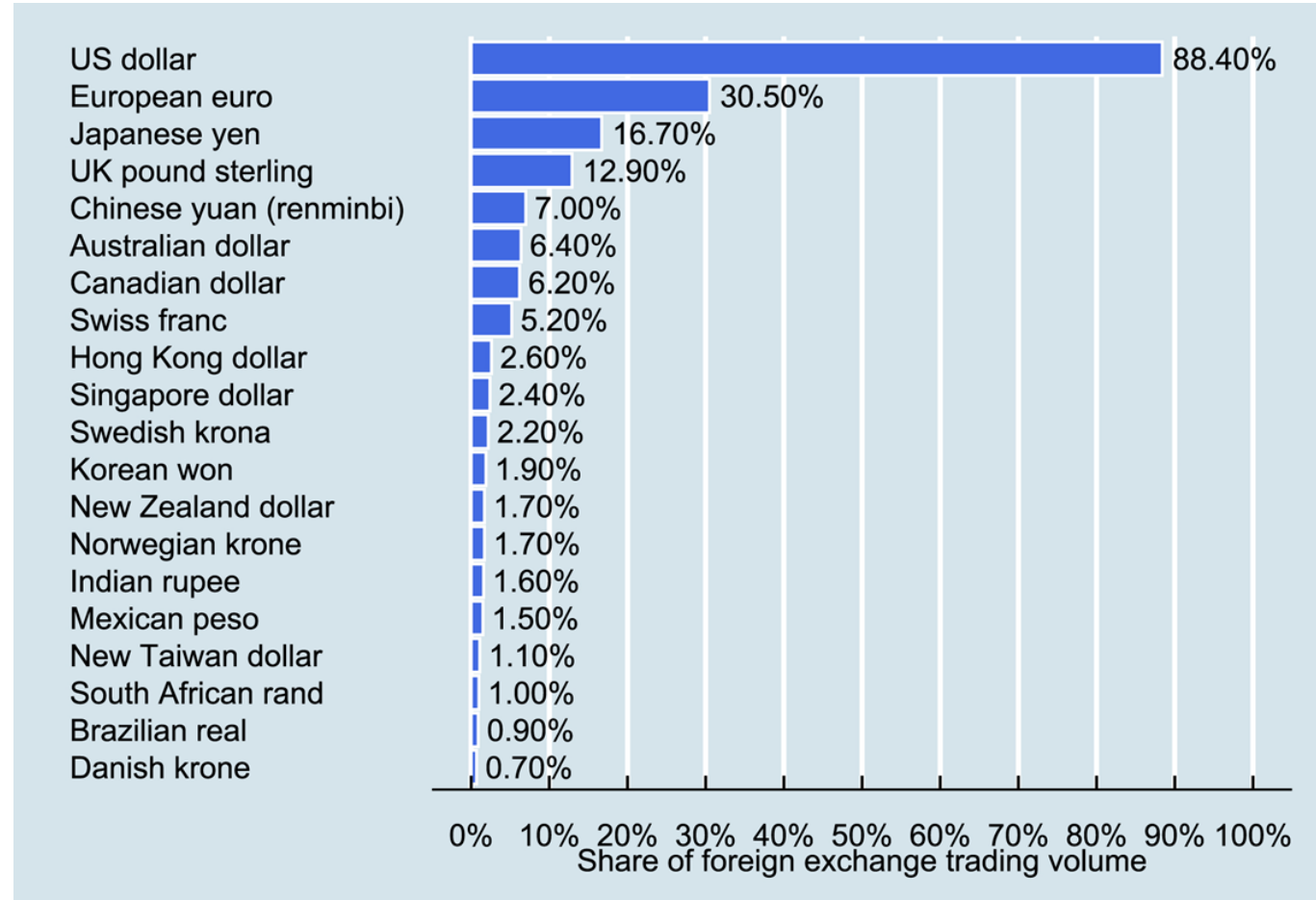
- Number of quotes on Bloomberg in the EUR/USD trading pair (May 26, 2023)
- The “witching hour” when New York closes and before Tokyo opens is clearly visible.
- The witching hour is the final hour of trading before derivatives contracts expire. Traders often use terms such as “triple witching,” which refers to the expiration of stock options, index options, and index futures on the same day.



1.3 FX Global Exchange

Leading traded currencies

- The international role of the US\$ is unrivaled.
- Emerging market currencies are growing in importance, mostly to the detriment of other currencies.
- Before the euro, the Deutsche Mark (DEM) was #2 and the French Franc was #5.

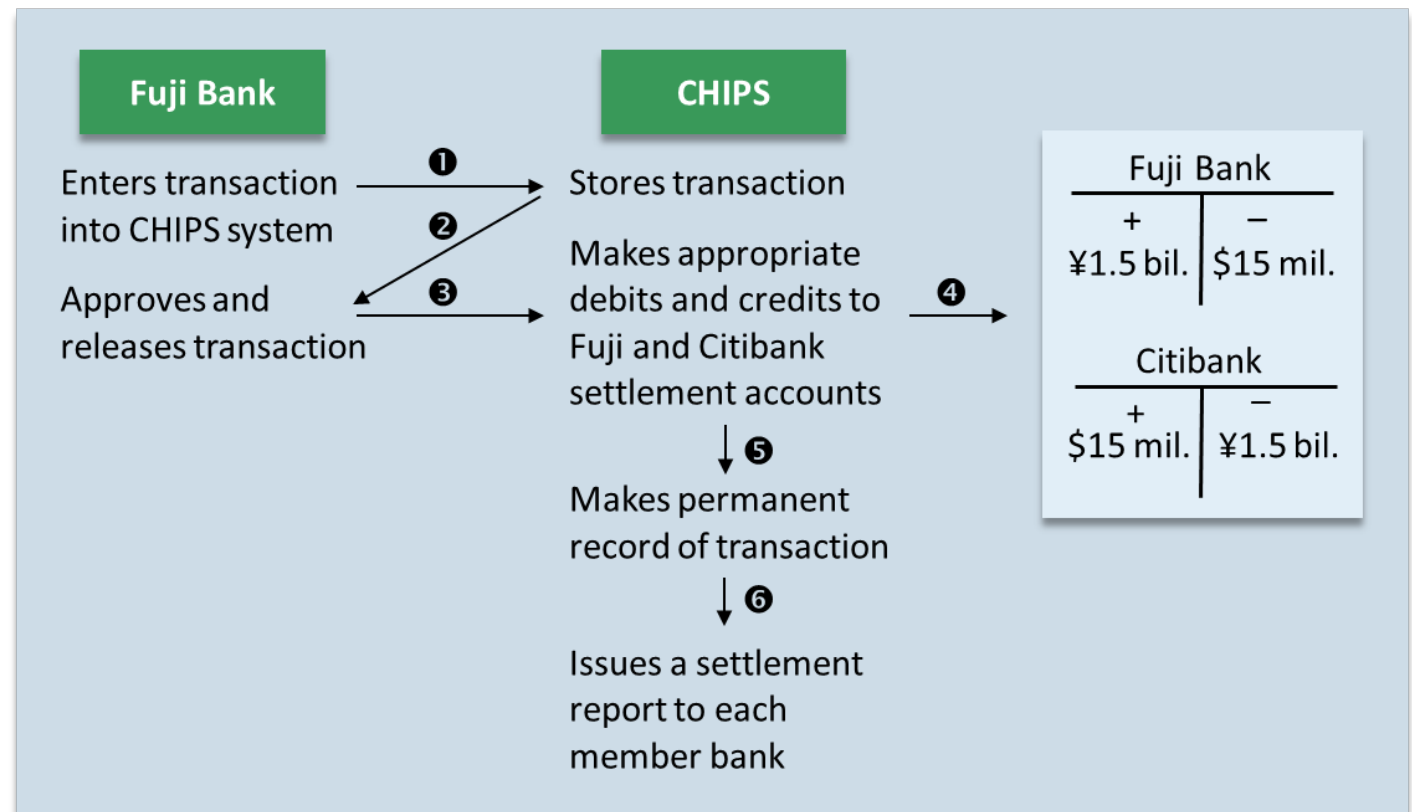


Global FX Exchange (4 of 4)

Clearing house

- In the US, where all foreign exchange transactions involving dollars are cleared, electronic funds transfers between banks are processed through the *Clearing House Interbank Payments System (CHIPS)*.
- Settlement payments are debited from and credited to a settlement account established by the New York Federal Reserve Bank for member banks.

Example: Fuji Bank sells \$15 million to Citibank for ¥1.5 billion



Spot Market

- Spot quotes
- Bid-ask spread
- Direct vs. indirect quotation and bid and ask prices
- Cross rates
- Currency arbitrage

Spot Quotes

- **Spot price:** the price quote of a currency for immediate delivery.
 - Widely disseminated in news outlets.
 - Two ways to quote a foreign currency:
 - **American terms:** the number of US dollars it takes to buy one unit of the foreign currency.
 - **European terms:** the number of foreign currency units it takes to buy one US dollar.

American terms

- \$/Foreign currency
- Direct quote in US
- Indirect quote outside US

European terms

- Foreign currency/\$
- Direct quote outside US
- Indirect quote in US

American Terms	American Terms	European Terms	European Terms
USD/EUR	Euro-Dollar	JPY/USD	Dollar-Yen
USD/AUD	Aussid-Dollar	CAD/USD	Dollar-Cad
USD/NZD	Kiwi-Dollar	SEK/USD	Dollar-Stockie
USD/GBP	Sterling or Cable	NOK/USD	Dollar-Nockie

Spot Quotes

The quotes for the JPY and the KRW are 2 digits after the decimal point and 4 digits for the other listed currencies.

Currency quotes on August 26, 2020

Currency	American Terms	European Terms
Euro (EUR)	1.1815	
Japanese Yen (JPY)		106.25
British Pound (GBP)	1.3188	
Swiss Franc (CHF)		0.9090
Australian Dollar (AUD)	0.7222	
Russian Rubble (RUB)		75.6160
Mexican Peso (MXN)		21.8840
Danish Krone (DKK)		6.2985
Hong Kong Dollar (HKD)		7.7503
South Korean Won (KRW)		1186.48

Pip

- The **last digit of a quote** represents a point, or **“pip”** (point in percentage).
 - For currencies quoted **4 digits** after the decimal point, a pip is an increment of **0.0001**.
 - For currencies quoted **2 digits** after the decimal point, a pip is an increment of **0.01**.
- A “pipette” is a $1/10^{\text{th}}$ of a pip.
 - The euro could be quoted at 1.1815**6**. The last digit in that case represents a “pipette.”

pip

Currency	Quote	Pip
Euro (EUR)	1.181 5	0.000 1
British Pound (GBP)	1.318 8	
Japanese Yen (JPY)	106.2 5	0.0 1
South Korean Won (KRW)	1186.4 8	

Pip

Traders refer to the number up to two digits after the decimal point as the “big figure.”

When traders communicate, they often omit the big figure and quote the currency in pips.

However, any traders not knowing the big figure will not stay in business for long

1.18156 USD/EUR

The diagram illustrates the components of the currency pair 1.18156 USD/EUR. The number 1.18 is red and labeled 'The Big Figure' with a red bracket. The number 15 is green and labeled '15 pips' with a green bracket. The number 6 is blue and labeled '6 pipettes' with a blue bracket.

Bid-Ask Spread

- Currency prices are different for buyers and sellers and are subjected to different prices.
 - Buy at the **bid price**, (i.e., buy the *base currency*)
 - Sell at the **ask price**, (i.e., sell the *base currency*). Ask price > bid price
- If the *dealer* is quoting pound sterling (£) at \$1.3183-88, they are willing to buy the currency at \$1.3183 and sell it for \$1.3188. We must buy at \$1.3188 and sell at \$1.3183.
- The percentage spread is $\frac{\text{Ask price} - \text{Bid price}}{\text{Ask price}} \times 100$

Example:

$$\frac{1.3188 - 1.3183}{1.3188} \times 100 = 0.038\%$$

1.3 Global FX Exchange

- Banker in Paris, France contacts a banker in Warsaw, Poland and asks how many zloty for 15 million euros. (we don't know if he is buying or selling).
- The trader in Warsaw answers back: 65/70 (bid (buy)/ask (sell)).
- On that day the bid (buy) was 4.31**65** and the ask (sell) was 4.31**70**
- The French banker reveals he wishes to buy 15 million euros.
- The Polish trader doesn't waste time and confirms the trade.
- These trades are then relayed to the back office for payment.

Example of a trade conversation

```
END FROM XXAH 1
# EURPLN 15 MIO
# 65/70
# I BUY
# TO CONFIRM 15 MIO AGREED AT 4.3170 I SELL EUR
# VAL 22JUN2018
# MY PLN TO TR BANK WARSAW ACC 1234567
# THANKS AND BYE
# TO CONFIRM AT 4.3170 I BUY 15 MIO EUR
# VAL 22JUN2018
# MY EUR TO TR BANK PARIS ACC 9876543
# THANKS AND BYE
#
# #END LOCAL#
^ ##ENDED AT 09:42 GMT##
```

Cross Rates

1. A cross rate is the exchange rate between two currencies calculated using a third currency. It is used when the two currencies aren't commonly traded directly against each other.
2. How are cross rates calculated?
3. Compare the exchange rate of each currency in the pair against a third currency, often the US dollar
2. Use the exchange rates against the third currency to calculate the exchange rate between the two currencies

Cross rates

1. Examples of cross rates:

- EUR/AUD (the exchange rate between the euro and the Australian dollar)
- EUR/GBP (the exchange rate between the euro and the British pound)
- CHF/JPY (the exchange rate between the Swiss franc and the Japanese yen)
- AUD/NZD (the exchange rate between the Australian dollar and the New Zealand dollar)

Cross Rates

1. Most currencies are quoted against the US \$.
2. But what if you want to purchase one foreign currency with another foreign currency, e.g., the Brazilian real and Thai baht?
3. We need to calculate the **cross rate** between the real and baht.
 - **Bid cross rate** – buying baht with reais (plural of real)
 - **Ask cross rate** – buying reais with baht

Cross Rates

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The Direct bid rate for the real in Bangkok can be found by recognising that selling reais (plural of real) in exchange for Baht is equivalent to combining two transactions

Bid and Ask rates for Real and Baht:

1. Brazilian real = R0.9955-1.0076/\$
2. Thai baht = B25.2513-3986/\$

Cross Rates

Bid and Ask rates for Real and Baht:

Brazilian real = **Bid/buy**-R0.9955- **Ask/sell**-1.0076/\$

Thai baht = **Bid/buy**-B25.2513-**Ask/sell**- B25.3986/\$

1. Selling the Real for dollars (which is the same as buying dollars with reais)
the asking rate of R 1.0076/US\$ and
3. Selling those dollars for baht at the bid rate of B25.2513/US \$.

These transactions result in the bid cross rate for the Real being the bid rate for the Baht divided by the asking rate for the Real, as shown in the next PPT slide

Cross Rates

- Most currencies are quoted against the US \$.
- But what if you want to purchase one foreign currency with another foreign currency, e.g., the Brazilian real and Thai baht?
- As stated above, we need to calculate the **cross rate** between the real and baht.
 - **Bid cross rate** – buying baht with reais (plural of real)
 - **Ask cross rate** – buying reais with baht

Example

- Brazilian real = R0.9955-1.0076/\$
- Thai baht = B25.2513-3986/\$

$$\begin{aligned}\text{Bid cross rate} &= \frac{\text{Bid rate for Thai baht} / \$}{\text{Ask rate for Brazilian real} / \$} \\ &= \frac{\text{B25.2513}}{\text{R1.0076}} = \text{B25.0608} / \text{R}\end{aligned}$$

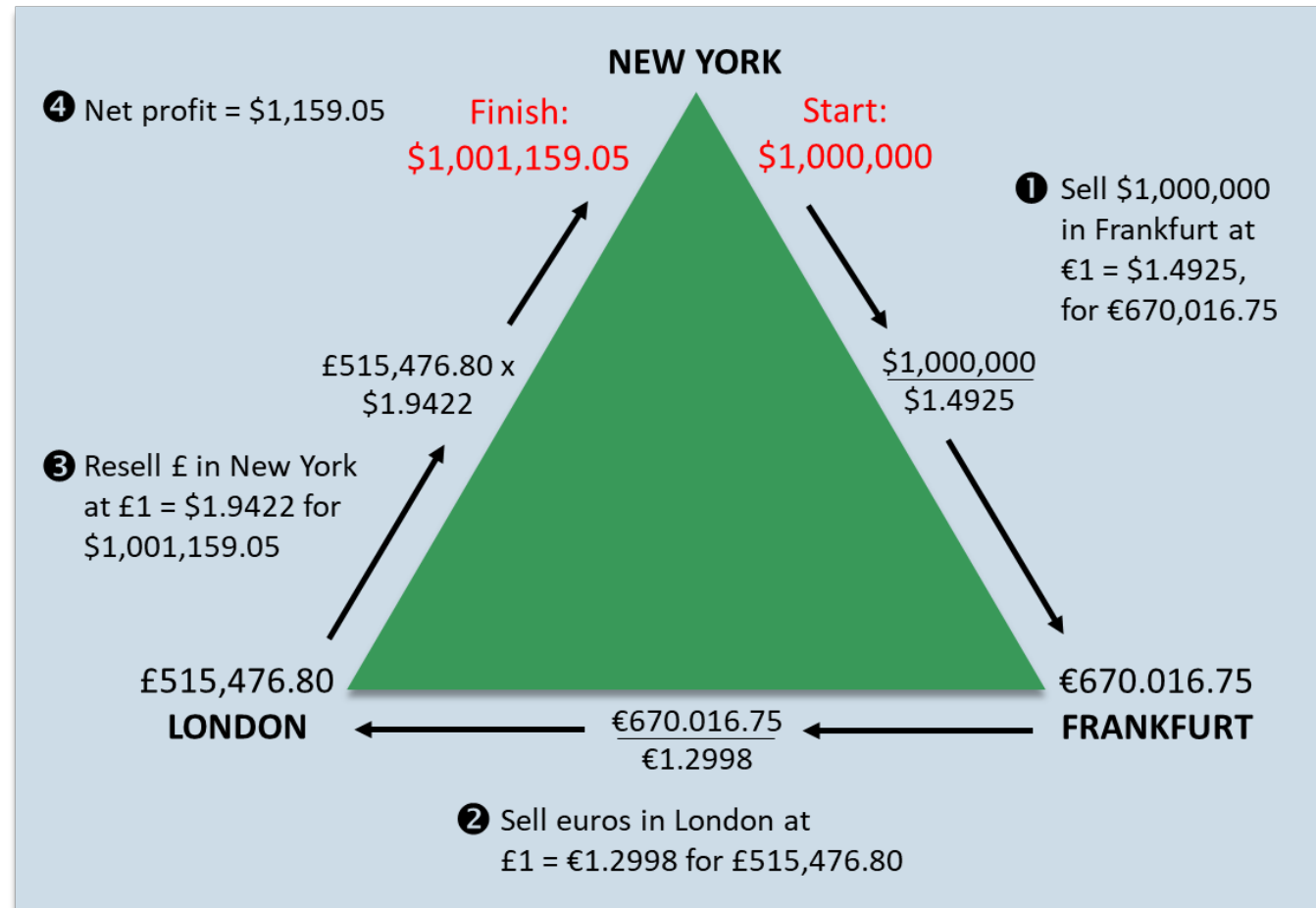
$$\begin{aligned}\text{Ask cross rate} &= \frac{\text{Ask rate for Thai baht} / \$}{\text{Bid rate for Brazilian real} / \$} \\ &= \frac{\text{B25.3986}}{\text{R0.9955}} = \text{B25.5134} / \text{R}\end{aligned}$$

Currency Arbitrage

- Arbitrageurs enforce the law of one price so prices of currencies in different markets are equivalent to each other.
- Suppose:
 - £ is bid at \$1.9422 in New York,
 - € is offered (asked) at \$1.4925 in Frankfurt, and
 - £ is offered at €1.2998 in London.
- The €/£ cross rate ($\$1.9422/\$1.4925 = €1.3013/£$) creates an arbitrage opportunity.

Currency Arbitrage (2 of 2)

- The trader would sell dollars for euros in Frankfurt, use the euros to acquire pounds in London, and sell the pounds in New York for dollars.
- This is called **triangular currency arbitrage**.



Forward Market

- What is a forward contract?
- Hedging a future payment with a forward contract
- Forward quotation

What is a forward contract?

- A contract that calls for delivery of a specified amount of one currency against dollar payment at a fixed future date, where the exchange rate is fixed at the time the contract is entered into.
- Normally available for 30-, 60-, 90-, 180-, or 360-day delivery, but bespoke (custom) maturities are possible.
- Used by multinationals to hedge against adverse currency movements.
 - Losses and gains from exchange-rate transactions result from the immediate adjustment of quotes in response to new political and economic information affecting exchange rates.
 - I.e., exchange-rate transactions are inherently risky.

Hedging a Future Payment with a Forward Contract

Example

- A US company buys computer chips from England and will owe £1 million in 90 days.
- The current spot price is \$1.25/£.
- The US company is **short** pounds – i.e., it owes pounds for future delivery.
- The company faces exchange-rate risk – if the spot rate goes up in 90 days, the dollar cost of the chips will go up, and vice versa.
- To hedge against exchange-rate risk, the company buys a 90-day forward contract that trades at \$1.26/£.
 - Locks in the rate of \$1.26/£.
 - If the spot rate in 90 days > \$1.26/£, the company has gained from the contract; if not, the company has lost from it.
 - Either way, the company that bought the forward contract can plan to pay \$1,260,000 regardless.

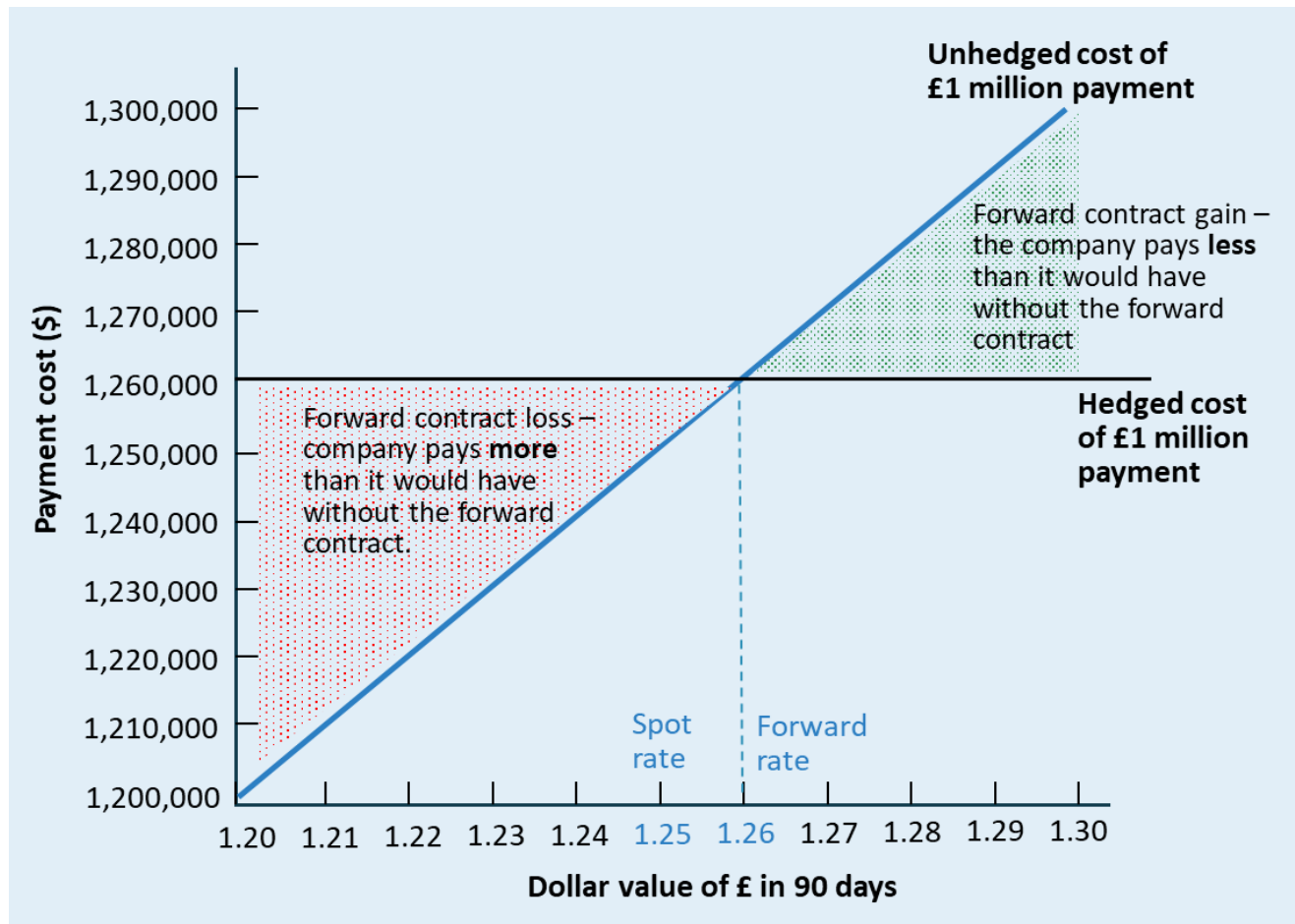
Hedging a Future Payment with a Forward Contract

1. The company faces exchange-rate risk – if the spot rate goes up in 90 days, the dollar cost of the chips will go up, and vice versa.
2. The importer is offsetting a short position in Pounds by going **long** in the forward market- That is, by buying pounds for future delivery.
3. The use of the forward contract enables the importer to convert a short underlying position in Pounds to a zero net exposed position, with the forward contract receipt of one million Pounds cancelling out the amount of one million Pounds and leaving the importer with a net liability of \$1,260,000.

Hedging a Future Payment with a Forward Contract

3 key points

1. The gain or loss on the forward contract is unrelated to the current spot rate of \$1.25.
2. The forward contract gain or loss exactly offsets the change in the dollar cost of the chip order that is associated with movements in the pound's value.
3. The bank must deliver the pounds, and the importer must buy them at the prearranged price.



Comments

1. Gains or losses on the forward contract are unrelated to the current spot rate of \$1.25.
2. The forward contract gain or loss exactly offsets the change in the dollar cost of the textile order that is associated with movements in the Pound's value. For example, if the spot price of the Pound in 90 days is \$1.20, the importer's cost of delivery is \$1.29 million. However, the forward contract has a gain of \$30,000 (the difference between the exchange rate of \$1.29 vs \$1.26 for a one million Pound commitment).

Forward Quotation

- Forward prices can be quoted in two ways:

Outright rate

- Actual prices quoted to commercial customers

Swap rate

- The difference between the forward rate and the spot rate
- Quoted in the interbank market

- Swap rate example

- The A U D/U S D pair is trading at \$0.7231/A\$ (spot rate).
- The 90-day forward contract on the A U D/U S D is trading at \$0.7233/A\$.
- The outright rate for the forward is \$0.7233/A\$.

Swap rate = \$0.7233 – \$0.7231, or 2 pips

Forward rate at Premium or Discount

1. Forward prices can be quoted in two ways: Outright and Swap rate

2. If forward/outright rate $>$ spot rate

Then forward contract trades is at a **premium**

3. If forward/outright rate $<$ spot rate

Then, forward contract trades is at a **discount**

Forward Quotation

- Points can be positive or negative

Positive points

- Outright rate > spot rate
- Forward contract trades at a **premium**

Negative points

- Outright rate < spot rate
- Forward contract trades at a **discount**

Forward premium or discount

$$\frac{\text{Forward rate} - \text{Spot rate}}{\text{Spot rate}} \times \frac{360}{\text{Contract days}}$$

AUD/USD example

$$\text{Forward premium} = \frac{0.7233 - 0.7231}{0.7231} \times \frac{360}{90} = 0.11\%$$